

In the given experiment we needed to find a value to for gravity. We accomplished this by dropping a pencil from 7 ft and measuring the time it took to hit the ground. We chose to have Nick stand on a chair, Cole records the time, and Josh record the drop. We used this strategy because we had three people, and it was easy to have everyone play a part.

We used a tape measure to find our height then an iPhone stopwatch to record the time. We then utilized the slow motion from the recording to determine an accurate time. We used three trials in effort to increase accuracy. Along with extra trials we chose to do it inside to avoid the wind as well as other weather conditions. For our first pencil drop, we measured our time to be 0.66 seconds. The next times that we dropped the pencil we got 0.60 and 0.62 seconds, respectively. We then plugged the average of these three values, 0.63, into the equation to find our gravity to be 10.75 m/s^2 , which ends up being about 0.94 off. We then compared our time to what it should have been with our initial height and seen that the time should have been about 0.66 seconds. So, we had some error with the

second and third trials, but our first measurement was spot on.

Handwritten calculations on lined paper:

$$t = 0.63s \quad p_0 = 2.1336m \quad p_f = 0$$

$$p(0.63) = 0 = 2.1336 - \frac{a(0.63)^2}{2}$$

$$a = \frac{2.1336 \cdot 2}{(0.63)^2} = 10.75 m/s^2$$

Handwritten calculations on lined paper:

$$2.1336 = \frac{1}{2} (9.81) t^2$$

$$t = 0.660$$

We deemed these to be respectably close considering some of our errors. Our first error, looking back at the recording, we saw that we had the measuring tape slanted for the first trial which will impact the equation since our initial height is now different. Another reason that we might not have gotten exactly $9.81 m/s^2$ is because we might not have started the stopwatch at the exact same time that Nick dropped the pencil. One minute reason that we might have not gotten exactly specific gravity is that we are about 1000 feet above sea level which will impact our calculations. All in all, though, it's a matter of milliseconds and it was probably some instrumental error.

Video Link:

<https://youtu.be/1fBTRNvbwVE?si=TMv1MZ9qHRN1HwM->

Group Photo:

